

Jeffrey Mei, Ph. D.

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WORK/RESEARCH EXPERIENCE

Staff Applied Scientist, Personalization

January 2023 –

SiriusXM Radio Inc, New York, NY 10020, USA (Remote)

- Implementing transformer models for next-song recommendation using feedback from 20M users
- Increased song completion rate by **0.8%**; improved song diversity on stations by **1.8%**; increased users' song thumb-up rate by **2.4%** ([UMAP '24](#) and [RecSys '25](#) publications)

Senior Machine Learning Scientist

December 2020 – January 2023

Wayfair LLC, Boston MA 02116, USA

- Developed transformer model with 30M customers' browsing history for next-item sequential recommendation, increasing order rates by **1%** and annual gross revenue by **\$67M** ([RecSys '22 publication](#) + workshop [keynote](#)) (Airflow, Google Cloud Platform, PyTorch, git, Docker)
- Created [interactive tool](#) (bokeh) to interpret and visualize learned item embeddings, and demonstrated that the model was implicitly learning product style, material, price and popularity attributes
- **Doubled** personalization rate by expanding model to transfer its learned attributes (e.g. style, material, functionality, price) across different furniture categories
- Implemented distributed multi-GPU training (PyTorch Lightning), reducing training time by **75%**

SELECTED FIRST-AUTHOR PUBLICATIONS AND INVITED TALKS

- "Semantic IDs for Music Recommendation" (RecSys '25) [10.1145/3705328.3748139](https://arxiv.org/abs/10.1145/3705328.3748139), 2025.
- "Negative Feedback For Music Personalization" (UMAP '24) [10.1145/3627043.3659553](https://arxiv.org/abs/10.1145/3627043.3659553), 2024.
- "Station and Track Attribute-Aware Music Personalization" (RecSys '23) [10.1145/3604915.3610239](https://arxiv.org/abs/10.1145/3604915.3610239), 2023.
- "A Lightweight Transformer for Next-Item Product Recommendation" (RecSys '22) [10.1145/3523227.3547491](https://arxiv.org/abs/10.1145/3523227.3547491), 2022.
- Industry Keynote talk, **Context-Aware Recommender Systems (CARS) Workshop**, 2022 <https://youtu.be/knziPzqBdfQ> (starts at 1:33:55)

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

June 2015 – September 2020

Woods Hole Oceanographic Institution, Woods Hole, MA

Ph. D., Mechanical Engineering (major: Oceanographic Engineering; minor: Data Science)

- [Dissertation](#): "Morphological Approaches To Understanding Antarctic Sea Ice Thickness" ([video](#))

New York University Abu Dhabi, Abu Dhabi, United Arab Emirates

August 2011 – May 2015

B. S. cum laude, Physics and Mathematics.

GPA: 3.8/4.0

- New York University Honors Scholar, 2015

- Semester study abroad at NYU Berlin/Humboldt-Universität zu Berlin (Germany), Spring 2013

OTHER SKILLS

Fluent in English, Mandarin Chinese; conversant in German (B2/C1), Russian (B1), Japanese (A2)
Experienced with Python (sklearn/pytorch/pandas), SQL, Spark, OpenCV, Linux/Unix, bash, git, GCP (Google Cloud Platform), Docker, AWS (Amazon Web Services), Apache Airflow, Presto, Databricks